

In[153]:=

```

P := {(* Reduced Markov chain for calculating hitting times from A, C, D, E, to B. *)
      (* AD,          DC,
      AB,          AC,          DA,
      EC,          DB,          AF,          ED *)
      (* AD *) {(n/2 - k - 1)/(n/2 - 1), 0,
                0, 0, 0,
                0, 0, 0, (k - 1)/(n/2 - 1)},
      (* DC *) {0, 0, 0, (n/2 - k)/(n/2),
                0, (k - 1)/(n/2), 0, 0},
      (* AB *) {0, 0, (n/2 - k - 1)/(n/2 - 1), 0,
                0, 0, k/(n/2 - 1), 0},
      (* AC *) {0, 1/(n/2 - 1), 0, (n/2 - k - 1)/(n/2 - 1),
                0, (k - 1)/(n/2 - 1), 0, 0},
      (* DA *) {0, 0, 0,
                0, 0, (k - 1)/(n/2),
                0, 1/(n/2), (n/2 - k - 1)/(n/2), 0},
      (* EC *) {0, 1/(n/2), 0, (n/2 - k)/(n/2),
                0, (k - 2)/(n/2), 0, 0, 1/(n/2)},
      (* DB *) {0, 0, (n/2 - k)/(n/2), 0,
                1/(n/2), 0, (k - 1)/(n/2), 0},
      (* AF *) {0, 0, k/(n/2 - 1),
                0, 0, (n/2 - k - 2)/(n/2 - 1), 0},
      (* ED *) {(n/2 - k)/(n/2), 0,
                0, 0, 0,
                1/(n/2), 0, 0, (k - 2)/(n/2)}
};

```

```

Psmall := P[[{1, 2, 3, 4, 5, 7, 8}, {1, 2, 3, 4, 5, 7, 8}]]
Hsmall := Simplify[Inverse[IdentityMatrix[7] - Psmall].{{1}, {1}, {1}, {1}, {1}, {1}, {1}}]
Hsmall // MatrixForm

```

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Simplify[P] // MatrixForm
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```
N1 := IdentityMatrix[9] - P
```

Simplify[N1] // MatrixForm

H := Simplify[Inverse[N1].{{1}, {1}, {1}, {1}, {1}, {1}, {1}, {1}, {1}}]

H // MatrixForm

Out[156]//MatrixForm=

$$\begin{pmatrix} \frac{-2+n}{2k} \\ \frac{4k+(-2+n)n}{-2n+2k(2+n)} \\ -1 + \frac{4}{n} + \frac{4k}{n} + n + \frac{-8+2n+n^2}{4k} \\ \frac{n^2}{-2n+2k(2+n)} \\ -2 + \frac{-2+n}{2k} + \frac{4}{n} + \frac{4k}{n} + n \\ -1 + \frac{4}{n} + \frac{4k}{n} + n + \frac{-4+n^2}{4k} \\ -2 + \frac{4k}{n} + n \end{pmatrix}$$

Out[157]//MatrixForm=

$$\begin{pmatrix} \frac{-2-2k+n}{-2+n} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2(-1+k)}{-2+n} \\ 0 & 0 & 0 & 1 - \frac{2k}{n} & 0 & \frac{2(-1+k)}{n} & 0 & 0 & 0 \\ 0 & 0 & \frac{-2-2k+n}{-2+n} & 0 & 0 & 0 & \frac{2k}{-2+n} & 0 & 0 \\ 0 & \frac{2}{-2+n} & 0 & \frac{-2-2k+n}{-2+n} & 0 & \frac{2(-1+k)}{-2+n} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2(-1+k)}{n} & 0 & \frac{2}{n} & \frac{-2-2k+n}{n} & 0 \\ 0 & \frac{2}{n} & 0 & 1 - \frac{2k}{n} & 0 & \frac{2(-2+k)}{n} & 0 & 0 & \frac{2}{n} \\ 0 & 0 & 1 - \frac{2k}{n} & 0 & \frac{2}{n} & 0 & \frac{2(-1+k)}{n} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2k}{-2+n} & 0 & 0 & \frac{-4-2k+n}{-2+n} & 0 \\ 1 - \frac{2k}{n} & 0 & 0 & 0 & 0 & \frac{2}{n} & 0 & 0 & \frac{2(-2+k)}{n} \end{pmatrix}$$

Out[159]//MatrixForm=

$$\begin{pmatrix} \frac{2k}{-2+n} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2(-1+k)}{-2+n} \\ 0 & 1 & 0 & -1 + \frac{2k}{n} & 0 & \frac{2-2k}{n} & 0 & 0 & 0 \\ 0 & 0 & \frac{2k}{-2+n} & 0 & 0 & 0 & -\frac{2k}{-2+n} & 0 & 0 \\ 0 & -\frac{2}{-2+n} & 0 & \frac{2k}{-2+n} & 0 & -\frac{2(-1+k)}{-2+n} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2-2k+n}{n} & 0 & -\frac{2}{n} & \frac{2+2k-n}{n} & 0 \\ 0 & -\frac{2}{n} & 0 & -1 + \frac{2k}{n} & 0 & \frac{4-2k+n}{n} & 0 & 0 & -\frac{2}{n} \\ 0 & 0 & -1 + \frac{2k}{n} & 0 & -\frac{2}{n} & 0 & \frac{2-2k+n}{n} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2k}{-2+n} & 0 & 0 & \frac{2(1+k)}{-2+n} & 0 \\ -1 + \frac{2k}{n} & 0 & 0 & 0 & 0 & -\frac{2}{n} & 0 & 0 & \frac{4-2k+n}{n} \end{pmatrix}$$

Out[161]//MatrixForm=

$$\begin{pmatrix} \frac{-((-2+n)n^2)+8k^2(2+n)+2k(-8-8n+n^3)}{2kn(4+n)} \\ \frac{16k^2+4k(-1+n)n+(-2+n)n^2}{4k(4+n)} \\ -1 + \frac{4}{n} + \frac{4k}{n} + n + \frac{-8+2n+n^2}{4k} \\ \frac{16k^2(2+n)+n^2(-4+n^2)+4k(-8-4n+n^2+n^3)}{4kn(4+n)} \\ -2 + \frac{-2+n}{2k} + \frac{4}{n} + \frac{4k}{n} + n \\ \frac{(-2+n)n^3+16k^2(2+n)+4kn(-4+n+n^2)}{4kn(4+n)} \\ -1 + \frac{4}{n} + \frac{4k}{n} + n + \frac{-4+n^2}{4k} \\ -2 + \frac{4k}{n} + n \\ \frac{(2+n)(4k+(-2+n)n)}{n(4+n)} \end{pmatrix}$$

In[162]:=

```
H CliqueToSameBridge := H[[1]]
H BridgeToAdjacentBridge := H[[2]]
H CliqueToOtherClique := H[[3]]
H CliqueToOtherBridge := H[[4]]
H BridgeToSameClique := H[[5]]
H BridgeToOtherBridge := H[[6]]
H BridgeToOtherClique := H[[7]]
H CliqueToSameClique := H[[8]]
H BridgeToSameBridge := H[[9]]
```

In[171]:=

In[172]:=

In[173]:=

In[174]:=

```

CommuteCliqueToSameBridge := HBridgeToSameClique + HCliqueToSameBridge
CommuteCliqueToOtherBridge := HCliqueToOtherBridge + HBridgeToOtherClique
CommuteBridgeToSameBridge := 2 * HBridgeToSameBridge
CommuteBridgeToOtherBridge := 2 * HBridgeToOtherBridge
CommuteBridgeToAdjacentBridge := 2 * HBridgeToAdjacentBridge
CommuteCliqueToSameClique := 2 * HCliqueToSameClique
CommuteCliqueToOtherClique := 2 * HCliqueToOtherClique

daggernormbridge :=
  Sum[(CommuteCliqueToSameBridge - CommuteCliqueToOtherBridge)^2, {i, n/2 - k}] +
  Sum[(CommuteCliqueToSameBridge - CommuteCliqueToOtherBridge)^2, {i, n/2 - k}] +
  Sum[(CommuteBridgeToSameBridge - CommuteBridgeToOtherBridge)^2, {i, k - 1}] +
  Sum[(CommuteBridgeToSameBridge - CommuteBridgeToOtherBridge)^2, {i, k - 1}] +
    2 * CommuteBridgeToAdjacentBridge^2

Simplify[daggernormbridge]

```

```

vol := Simplify[2 * (n/2) * (n/2 - 1) + 2 * k]
vol

```

Out[182]=

$$\left\{ \frac{(4k + (-2 + n)n)^2 (-32k + 32k^2 + n(4 + n)^2)}{4k^2(4 + n)^2} \right\}$$

Out[184]=

$$2k + \frac{1}{2}(-2 + n)n$$



In[185]:=

```
Simplify[daggernormbridge / (4 * vol ^ 2)]
Simplify[daggernormbridge / (4 * vol ^ 2) /. {k -> 1}]
```

Out[185]=

$$\left\{ \frac{-32 k + 32 k^2 + n (4 + n)^2}{4 k^2 (4 + n)^2} \right\}$$

Out[186]=

$$\left\{ \frac{n}{4} \right\}$$

In[187]:=

```
Simplify[daggernormbridge / (4 * vol ^ 2) /. {k -> 100, n -> 200}]
```

Out[187]=

$$\left\{ \frac{3}{578} \right\}$$

In[188]:=

```

averagecommutebridge := (CommuteBridgeToAdjacentBridge +
    CommuteBridgeToSameBridge * (k - 1) +
    CommuteBridgeToOtherBridge * (k - 1) +
    CommuteCliqueToSameBridge * (n/2 - k) +
    CommuteCliqueToOtherBridge * (n/2 - k)) / n;

averagecommuteclique := (CommuteCliqueToSameBridge * k +
    CommuteCliqueToOtherBridge * k +
    CommuteCliqueToOtherClique * (n / 2 - k) +
    CommuteCliqueToSameClique * (n / 2 - k - 1)) / n;

Simplify[averagecommutebridge]
Simplify[averagecommuteclique]
Simplify[averagecommutebridge - averagecommuteclique]

```

```
Simplify[averagecommutebridge /. {n -> 8, k -> 3}]
```

```
Simplify[averagecommuteclique /. {n -> 8, k -> 3}]
```

Out[190]=

$$\left\{ \frac{-64 k^3 + 16 k^2 (-12 + 4 n + n^2) + n^2 (-16 + 2 n^2 + n^3) + 4 k n (32 - 12 n - n^2 + 2 n^3)}{4 k n^2 (4 + n)} \right\}$$

Out[191]=

$$\left\{ \frac{-64 k^3 + (-2 + n) n^2 (4 + n)^2 + 16 k^2 (-12 + 6 n + n^2) + 4 k n (40 - 12 n + n^2 + 2 n^3)}{4 k n^2 (4 + n)} \right\}$$

Out[192]=

$$\left\{ -\frac{8 k + 8 k^2 - 4 n + 2 k n^2 + n^3}{4 k n + k n^2} \right\}$$

Out[193]=

$$\left\{ \frac{355}{24} \right\}$$

Out[194]=

$$\left\{ \frac{145}{8} \right\}$$

In[195]:=

```

daggernormbridgeclique := ((CommuteCliqueToOtherClique - averagecommuteclique) +
  (averagecommutebridge - CommuteCliqueToOtherBridge)) ^ 2 * (n/2 - k) +
  ((CommuteCliqueToOtherBridge - averagecommuteclique) +
  (averagecommutebridge - CommuteBridgeToOtherBridge)) ^ 2 * (k - 1) +
  ((CommuteCliqueToOtherBridge - averagecommuteclique) +
  (averagecommutebridge - CommuteBridgeToAdjacentBridge)) ^ 2 * (1) +
  ((CommuteCliqueToSameBridge - averagecommuteclique) +
  (averagecommutebridge - CommuteBridgeToSameBridge)) ^ 2 * (k - 1) +
  ((CommuteCliqueToSameClique - averagecommuteclique) +
  (averagecommutebridge - CommuteCliqueToSameBridge)) ^ 2 * (n/2 - k - 1) +

  ((CommuteCliqueToSameBridge - averagecommuteclique) + (averagecommutebridge - 0)) ^ 2 *
  (1) +

  ((0 - averagecommuteclique) +
  (averagecommutebridge - CommuteCliqueToSameBridge)) ^ 2 * (1)

```

In[196]:=

```

Simplify[daggernormbridgeclique / (4 * vol ^ 2)]
Simplify[daggernormbridgeclique / (4 * vol ^ 2) /. {k -> 1}]
Simplify[daggernormbridgeclique / (4 * vol ^ 2) /. {n -> 10, k -> 2}]

```

Out[196]=

$$\left\{ \frac{16 k (2 + n) + n (4 + n)^2 + 8 k^2 (12 + 6 n + n^2)}{k^2 n^2 (4 + n)^2} \right\}$$

Out[197]=

$$\left\{ \frac{8 + n}{n^2} \right\}$$

Out[198]=

$$\left\{ \frac{981}{9800} \right\}$$

In[199]:=

```

daggernormclique := 2 * (CommuteCliqueToSameClique ^ 2)

```

```

Simplify[daggernormclique / (4 * vol ^ 2)]

```

Out[200]=

$$\left\{ \frac{8}{n^2} \right\}$$

In[201]:=

```

daggernormclique := 2 * (CommuteCliqueToSameClique ^ 2)
Simplify[daggernormclique]
Simplify[4 * vol ^ 2]
Simplify[daggernormclique / (4 * vol ^ 2)]

```

Out[202]=

$$\left\{ 8 \left(-2 + \frac{4k}{n} + n \right)^2 \right\}$$

Out[203]=

$$(4k + (-2 + n)n)^2$$

Out[204]=

$$\left\{ \frac{8}{n^2} \right\}$$

In[205]:=

```

daggernormbridgebridge := 2 * (CommuteBridgeToSameBridge ^ 2) +
  2 * (CommuteBridgeToAdjacentBridge - CommuteBridgeToOtherBridge) ^ 2
Simplify[daggernormbridgebridge / (4 * vol ^ 2)]

```

Out[206]=

$$\left\{ \frac{8(8 + 4n + n^2)}{n^2(4 + n)^2} \right\}$$

In[207]:=

```

Simplify[daggernormbridge / (4 * vol ^ 2)]

```

Out[207]=

$$\left\{ \frac{-32k + 32k^2 + n(4 + n)^2}{4k^2(4 + n)^2} \right\}$$

In[208]:=

```

Simplify[daggernormbridgebridge / (4 * vol ^ 2)]
Simplify[daggernormclique / (4 * vol ^ 2)]
Simplify[daggernormbridgeclique / (4 * vol ^ 2)]

```

Out[208]=

$$\left\{ \frac{8(8 + 4n + n^2)}{n^2(4 + n)^2} \right\}$$

Out[209]=

$$\left\{ \frac{8}{n^2} \right\}$$

Out[210]=

$$\left\{ \frac{16k(2 + n) + n(4 + n)^2 + 8k^2(12 + 6n + n^2)}{k^2n^2(4 + n)^2} \right\}$$

In[211]:=

```
FullSimplify[daggernormbridge > daggernormbridgebridge, k ≥ 1 && n ≥ 6]
FullSimplify[daggernormbridge > daggernormbridgeclique, k ≥ 1 && n ≥ 6]
FullSimplify[daggernormbridge > daggernormclique, k ≥ 1 && n ≥ 6]
FullSimplify[daggernormbridgeclique > daggernormclique, k ≥ 1 && n ≥ 6]
FullSimplify[daggernormbridgeclique > daggernormbridgebridge, k ≥ 1 && n ≥ 6]
FullSimplify[daggernormclique > daggernormbridgebridge, k ≥ 1 && n ≥ 6]
FullSimplify[daggernormbridge == Max[daggernormbridge,
    daggernormbridgeclique, daggernormbridgebridge], k ≥ 1 && n ≥ 6 && k ≤ n/2]
FullSimplify[daggernormbridge ≥ daggernormclique, k ≥ 1 && n ≥ 6 && k ≤ n/2]
```

Out[211]=

$$\left\{ -\frac{(4k + (-2 + n)n)^2 (32k^2 n^2 + 128k^2(2 + n) - n^3(4 + n)^2)}{4k^2 n^2 (4 + n)^2} \right\} > \{0\}$$

Out[212]=

$$\left\{ -\frac{(4k + (-2 + n)n)^2 (192k^2(2 + n) - n(4 + n)^2(-4 + n^2) + 32k(4 + n(2 + n)))}{4k^2 n^2 (4 + n)^2} \right\} > \{0\}$$

Out[213]=

$$\left\{ \frac{(4k + (-2 + n)n)^2 (32(-1 + k)k + n(4 + n)^2)}{4k^2 (4 + n)^2} \right\} > \left\{ 8 \left(-2 + \frac{4k}{n} + n \right)^2 \right\}$$

Out[214]=

$$\left\{ -\frac{(4k + (-2 + n)n)^2 (-16k(2 + n) + 16k^2(2 + n) - n(4 + n)^2)}{k^2 n^2 (4 + n)^2} \right\} > \{0\}$$

Out[215]=

$$\left\{ \frac{(4k + (-2 + n)n)^2 (16k(2 + n) + 16k^2(2 + n) + n(4 + n)^2)}{k^2 n^2 (4 + n)^2} \right\} > \{0\}$$

Out[216]=

$$\left\{ \frac{32(2 + n)(4k + (-2 + n)n)^2}{n^2 (4 + n)^2} \right\} > \{0\}$$

Out[217]=

True

Out[218]=

$$\left\{ \frac{(4k + (-2 + n)n)^2 (32(-1 + k)k + n(4 + n)^2)}{4k^2 (4 + n)^2} \right\} \geq \left\{ 8 \left(-2 + \frac{4k}{n} + n \right)^2 \right\}$$